# Workshop Pole Barn: Spec Quick Reference

Print this page and take it to every vendor call · 32 × 48 ft · 14 ft eave · gambrel · Prairie, Idaho · planned for 2028

> **Verify with the county and an Idaho engineer before ordering.** The snow, wind, frost, and seismic values below are Elmore County's published ones (site baselines); the rest are design requirements to confirm with the engineer.

## Design Loads (verify with the county)

| Parameter | Value | Notes |
|---|---|---|
| Ground snow load (Pg) | **120 psf** | The design load. The county's snow-load map puts Prairie in the 120 psf zone, where an Idaho engineer must design the building, and the county required a neighbor's building to be designed for it (owner, October 2026). |
| Flat-roof snow (approx.) | ≈ 84 psf | 0.7 × Pg. The truss designer calculates the real figure, with the reductions for the slopes. |
| Snow drift at the pitch break | Per engineering | Gambrels collect drift at the knuckle. |
| Basic wind speed (V) | **115 mph** | The county's residential figure (IRC); commercial uses 95 to 120 mph (IBC). ASCE 7, Risk Category II. |
| Wind exposure | **C** | Open ranch land. |
| Seismic | County zone C | Prairie is south of Featherville. The engineer sets the design category. |
| Frost | County 30 in zone | The county asks for footings "30 in below the frost line," which is **60 in** below grade. Footings are specified to 60 in below the pad surface. Confirm the wording. |
| Loft live load | 40 psf minimum (60 is better) | Spec to the truss plant. |
| Height (county) | Mid-gable 20′-4″ above the pad; ridge 25′-7″ | An accessory building may be no taller than the house or 24 ft, whichever is greater, measured to the mid-point of the gable: under by 44″ from the pad surface (about 22.0 ft if the county averages the gambrel's whole gable). The zone limit is 35 ft. The pole barn guide's usual 14 ft wall and 27 ft peak are met from the floor; the guide does not bind an engineered building (see County rules). |

## Pad and Foundation

| Item | Spec |
|---|---|
| The pad | About 56 × 42 ft, level, built of imported fill. The fill is almost 2 ft over the original ground on the north edge and 6 in to 1 ft on the south edge (owner, October 2026). The building is centered on it with 4 ft of pad at each end and 5 ft along the long sides. |
| Fill | Ask the engineer whether it was compacted in lifts. If that is not documented, compact and test it before the slab, or deepen the gravel base. Check the north edge, where the fill is thickest and the margin is only 5 ft. |
| Pier holes | **30 in Ø × 60 in deep** minimum, measured from the pad surface. They pass through the fill and bear on undisturbed native soil, or on fill the engineer accepts. About 3 ft of native soil under the north row, 4 to 4½ ft under the south row. |
| Footing pad | 30 in Ø × 12 in thick, 4,000 psi concrete, with an 18 in Ø tube pier above it, to the bracket. |
| Posts and brackets | 3-ply 2×8 laminated, 16 ft long, **wet-set on concrete piers** (Perma-Column or Sturdi-Wall Plus). 19 in all: 14 on the sidewalls, 3 on the west gable (between the corners), and 2 door jambs on the east gable. |
| Post spacing | **8 ft OC maximum** on the sidewalls. Layout 8-8-8-8-8-8 for the 48 ft length, so the 16 ft girts land on posts; the west gable posts are 8 ft apart. |
| Slab | **5 in**, 4,000 psi, #4 rebar at 16 in OC each way, with control joints at about 11 ft. It slopes toward the east door; the amount is open, because ¼ in/ft would drop 12 in over this length (see the overview's open questions). Isolate the slab from posts and brackets with expansion-joint filler or 15 lb felt (the county's pole barn guide). |
| Under the slab | 4 in compacted gravel, then a 10 mil vapor barrier, then 4 in rigid foam, on the compacted pad fill. The stack puts the floor about 1′-1″ above the pad surface. |

## Structure

| Item | Spec |
|---|---|
| Girts | 2×6 SPF #2, **laid flat on the outside face of the posts at 24 in OC**, continuous over the posts and screwed to each one. 7 rows on each wall; see Wall Framing below |
| Trusses | Pre-engineered gambrel attic truss, 32 ft span, **2 ft OC**, PE stamped for 120 psf ground snow: 25 in all (23 common and 2 gable-end). The loft stair opening, the loft doors, and the hoist beam go on the truss drawings. |
| Truss heel height | 12 in minimum (for insulation depth) |
| Truss carriers | (3) 1¾″ × 9¼″ LVL, side-bolted to each post with (3) ½ in galvanized through-bolts (Detail 7) |
| Purlins | 2×6 SPF, 16–24 in OC per the truss design |
| Splash board | 2×10 PT, screwed to the posts below the base girt, in the same layer as the girts |

| Item | Spec |
|---|---|
| Door header | (3) 2×12 LVL or a steel beam (PE-sized), 12 in deep at most: only 14″ is free above the door head |
| Hurricane ties | H2.5A or stronger, **2 per truss bearing** (100 in all) |

## Wall Framing (Girts)

| Item | Spec |
|---|---|
| Member | 2×6 SPF #2 **laid flat on the outside face of the posts**: 5½ in tall and 1½ in thick, so the wall is about 10″ thick from the steel to the inside face of the posts. The girts run continuously over the posts, and the sheathing goes over them (Detail 1) |
| Rows | **7 rows on every wall, 24 in OC**, with their bottom edges at 0′ (the floor line), 2′, 4′, 6′, 8′, 10′, and 12′ above the floor. The lowest is a **pressure-treated base girt** (13 in above the pad): it backs the lower edge of the sheathing, and the 2×10 PT splash board is screwed to the posts below it. The highest row stops less than 2 ft below the eave |
| Boards | **16 ft boards.** The posts are 8 ft apart, so a board spans two bays and ends on a post. Nothing is cut off, and nothing is spliced anywhere else |
| Staggered joints | Rows alternate between two patterns so that neighboring rows never end on the same post: **16-16-16** on the north and south walls (16-16 on the west wall), and **8-16-16-8** (8-16-8 on the west wall), where each pair of 8 ft pieces is one 16 ft board cut in half. Every row takes the same number of boards, so the stagger costs nothing |
| 18 ft boards | Not on the sidewalls: an 18 ft board would end 2 ft past a post, where it cannot be fastened, so 2 ft would be cut off every board. They are used on the east wall, where the door jambs leave a 9 ft run on each side of the door in every row: one 18 ft board cut in half makes both pieces |
| Fasteners | 3½ in structural screws: **3 at each post a board crosses and 2 at each board end** (about 457 in all). The county's minimum is four 20d nails per girt at each post; the engineer confirms the screw schedule |
| Openings | At a sidewall window the 6′ row is cut; the 8′ row is the header, and a 2×6 sill and two jack studs frame the opening. At the walk door the base girt and the 2′, 4′, and 6′ rows stop at jack studs beside the door. At the east door every row stops at the jamb posts. The cut-out pieces make the sills and jacks |
| Corners | The sidewall girts run to the outside corner and the end-wall girts butt against them; the engineer details the corner |
| To order | 48 boards of 2×6×16 SPF #2 for the north, south, and west walls; 6 of 2×6×18 SPF #2 for the east wall; 10 of 2×6×16 PT for the base girts (materials, phase 6) |

## Envelope

### Wall assembly (outside → inside)

| Layer | Spec |
|---|---|
| 1. Siding | 26 ga steel, vertical panels, **barn red** |
| 2. Rainscreen | **1×4 PT furring at 24 in OC, vertical** (critical: do not skip) |
| 3. WRB | Tyvek CommercialWrap, taped seams |
| 4. Sheathing | **½ in OSB**, APA Exposure 1, 4×8 sheets laid horizontally with their long edges on the center lines of the 0′, 4′, 8′, and 12′ girt rows, so every edge is nailed to a girt; 8d ring-shank at 6 in on the edges and 12 in in the field (4 in at corners) |
| 5. Girts | 2×6 laid flat on the posts, at 24 in OC, screwed to each post (see Wall Framing) |
| 6. Posts | 3-ply laminated at 8 ft OC |

### Roof assembly (outside → inside)

| Layer | Spec |
|---|---|
| 1. Roofing | 26 ga steel with **factory Dripstop**, **beige** |
| 2. Underlayment | Synthetic (Titanium UDL30 or equal) over the entire roof |
| 3. Ice and water shield | FULL coverage on the lower (steep) slope; at the pitch break and around penetrations on the upper slope |
| 4. Deck | **⅝ in OSB** with H-clips between the purlins |
| 5. Purlins | 2×6 per the truss design |
| 6. Trusses | Gambrel attic, 32 ft at 2 ft OC |
| Snow guards | On the lower slope over the walk door, and over any other door under a roof edge. The east gable has no roof slope above it, so it needs none. The engineer or the steel supplier lays out the rows for the snow zone. |

## Openings

| Opening | Spec | Location |
|---|---|---|
| Overhead door | 14′ × 12′ insulated, R-12 or better, **wind rated for 115 mph**, low-headroom track (12 in above the door; the loft floor leaves 14″). Clears the 10′-11″ camper by 1′-1″ | Centered on the EAST gable |
| Loft double doors | 5′ × 6′ insulated double doors (French or barn-door style), upper lites optional | EAST gable, **centered above the main door at loft-floor level** |
| Landing deck | 14′ wide × 10′ deep (matches the main door), free-standing: 6×6 PT posts to grade piers, doubled 2×10 PT beams east-west, 2×10 PT joists at 12 in OC north-south, 5/4 or 2×6 PT decking, 42″ rail, diagonal knee braces | Free-standing off the east gable at the loft doors |
| Walk door | 3′ × 7′ insulated steel, pre-hung | SOUTH sidewall, near the east end |
| Sidewall windows | 4′ × 3′ (wide × tall) insulated vinyl sliders, low-E glass, **sill 5′ above the floor**, DP-30 or better (ask the engineer for the rating at 115 mph). Nail-fin frames with jamb extensions that reach the inside face of the posts (9¼″ behind the sheathing). | NORTH wall: 3, centered in bays 2, 4, and 6, counting from the west end. SOUTH wall: 3, in bays 3, 4, and 5, clear of the stair and the walk door. |
| Loft window | 3′ × 4′ (wide × tall) insulated vinyl single-hung, low-E glass, **sill 3′ above the loft floor** | WEST gable, centered, at the loft |

### Window pre-planning

- Each window is centered **between two posts**, so no post is moved. At each sidewall window cut the girt row that crosses it (the 6′ row): the girt at 8′ is the header, so frame the sides with 2×6 jack studs and add a 2×6 sill and blocking for the J-trim (see Wall Framing).
- Flash every window the same way: **sill pan first, then the jamb tape, then the head tape**, lapped under the housewrap above and over it below. The rainscreen cavity stays open behind the trim, and the steel is finished around each window with J-trim and a head flashing.

### Loft door and deck pre-planning

- Tell the truss plant the **loft door rough opening** on the east gable: about 5′-6″ wide × 6′-6″ tall, centered, bottom at the loft floor.
- Header above the loft doors: (2) 2×10 LVL, or as the engineer says.
- **Hoist beam included:** a W6 steel I-beam through the east gable, 21′-6″ long: 9 ft inside the loft and 12′-6″ out, which is 2 ft past the deck's outer edge. A load is lifted from the ground at the deck's edge, not through the deck, and comes in over the rail at loft-floor level. The engineer sizes it for a 500–1,000 lb working load. Anchor it to the truss top chord and to blocking inside the loft, and hang a trolley and a 1-ton chain fall with a 25 ft lift from it. A steel plate on the tip stops the trolley a few inches short of the end.
- **Hoist braces:** two raking 6×6 posts (or steel tube), about 11′-9″ each, run from the outer corners of the deck up to the tip of the beam. They meet the beam only at the tip, so the trolley never passes them, and the hook hangs between them: a load up to about 4 ft wide rises clear. A steel shoe on the end of each outer deck beam seats a brace, and the tip plate has an ear for each. The braces carry most of the tip load down into the deck, so the engineer sizes the beam, the tip plate, the braces, the deck beams and posts, and the piers together, and checks the deck for snow. The braces tie the deck to the hoist beam, so the engineer also checks the load path between the deck and the building in both directions. The rail is framed around the braces.
- Deck piers: 4 concrete piers below frost (60 in), 18 in Ø tubes with a belled footing. Anchor the 6×6 posts with post bases. The posts stand on the door-jamb lines, so they do not narrow the door.
- The deck is **free-standing** (no ledger on the building), which protects the rainscreen and the siding from leaks.
- The four posts need lateral bracing (14 ft wide and 14 ft tall is tall and narrow): diagonal knee braces or X-bracing per the engineer.
- Decking: PT, composite, or Trex, the owner's choice, with stainless or hot-dip galvanized fasteners.
- **No awning** over the deck. **No exterior stairs** to grade: the deck is reached from the loft only.

## Loft Stairs

| Item | Spec |
|---|---|
| Layout | L-shape with a 4′ × 4′ landing at the **southwest corner**, 6′-5″ up: the highest that leaves 6′-8″ of headroom under the loft floor (13′-2″ up) |
| First flight | SOUTH sidewall, east to west, 9′-2″ of run, rising 6′-5″ in 11 risers to the landing |
| Second flight | WEST gable wall, south to north, 11′-0″ of run, rising 7′-7″ in 13 risers to the loft floor |
| Width, rise, tread | 48 in clear minimum; 7 in rise and 11 in tread (target): 24 risers in all, and one fewer tread than risers in each flight |
| Rails and guard | Handrails on both sides, full length, 34–38 in; a guard of 42 in minimum around the loft opening |
| Stringers | 2×12 PT or LVL |
| Opening in the trusses | About 4′ × 11′ at the west wall, over the whole second flight, off the southwest landing: tell the truss plant when ordering (above the landing the stair has no headroom under the floor) |

## Mechanical and Electrical

| Item | Spec |
|---|---|
| Service | 200 A sub-panel. **The feed source is not settled:** the pump house (inverters and batteries) is about 29 ft south of the south wall, the solar container about 33 ft south, and the planned house about 54 ft to the southeast. Check that the source can supply the loads below. |
| Feeder conduit | 2 in PVC Schedule 40 underground, before backfill, entering at the panel |
| Welder | 240 V / 50 A dedicated |
| Air compressor | 240 V / 30 A dedicated |
| RV plug | 30 A TT-30R, near the RV bay |
| Mini-split | 240 V / 15–30 A per the condenser's nameplate |
| Wood stove blower | 120 V / 15 A near the stove |
| Lighting | LED high-bay, 4000–5000 K, 15,000+ lm each, on several circuits |
| Heat | Wood stove (primary) plus a mini-split heat pump |
| Cooling | The mini-split (same unit) |
| Plumbing and drains | NONE, and no floor drain |

## Colors

| Surface | Color |
|---|---|
| Roof steel | Beige |
| Wall steel | Barn red (dark, deep) |
| Trim | Beige (matches the roof) |
| Wainscot | None |

### What NOT to do

- ❌ Steel screwed straight to the OSB (no rainscreen): a sure condensation failure
- ❌ 15 or 30 lb felt behind the wall steel: it tears and traps moisture
- ❌ Poly vapor barrier outside the OSB: it traps moisture and rots the sheathing
- ❌ Foil-faced bubble wrap as the WRB: it is not a real WRB
- ❌ Ice and water shield on the walls: a vapor trap, the wrong tool
- ❌ Wood posts buried in the dirt at this snow load: use brackets on piers
- ❌ Skipping the H2.5A hurricane ties: 115 mph wind needs them
- ❌ Skipping the rebar in the slab: 5 in of concrete under stored loads will crack
- ❌ Pouring the slab or setting footings on uncompacted fill without the engineer's say-so

Mountain Prairie Acres · Workshop pole barn spec quick reference · updated October 23, 2026 · counts and sizes are filled in from `docs/workshop_model.js`.